

LESSON 5.6

SCALE FACTORS – AREA



LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can relate the areas of rectangles to their scale factors.
- I can relate the perimeters and areas of triangles to their scale factors.
- I can relate the perimeters and areas of trapezoids to their scale factors.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- What is the relationship between an original figure's area and its scale model's area?
- How does the scale factor between scaled figures relate to the areas of the figures?
- Do these relationships extend to other two-dimensional geometric figures?

LESSON NARRATIVE

Students continue their exploration of scale factor by determining the relationship between the area of scaled rectangles and the scale factor relating the figures. The prior lesson and today's lesson are synthesized for students in the 5.6.3 Guided Instruction, where students make connections between scale factor, perimeter, and area of other two-dimensional geometric figures.

COMPONENT SUMMARY

5.6.1 WARM-UP (~5 MIN.)

Determine a scale factor from the world's largest gummy bear to a regular gummy bear

5.6.2 EXPLORATION (15-20 MIN.)

Explore and determine the relationship between scale factor and area

5.6.3 GUIDED INSTRUCTION (15-20 MIN.)

Determine whether the relationships between scale factor, perimeter, and area apply to other geometric figures

5.6.4 WRAP-UP (~5 MIN.)

Reflect on mindset throughout today's lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Colored pencils, highlighters, or crayons

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make mistakes when determining scale factor if the order of their figures is incorrect.

THINGS TO CONSIDER

- Consider students' levels of understanding of the content over the last lessons to determine how much of component 5.6.3 Guided Instruction needs to be guided or include the teacher modeling versus released to the students.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

5.6.3 Guided Instruction engages students in a Notice and Wonder routine. Use this as an opportunity for mathematical discourse and to make connections to prior knowledge of finding the area of different figures. This will provide an entry point for students as they apply what they have learned about scale factor to new shapes in this section.

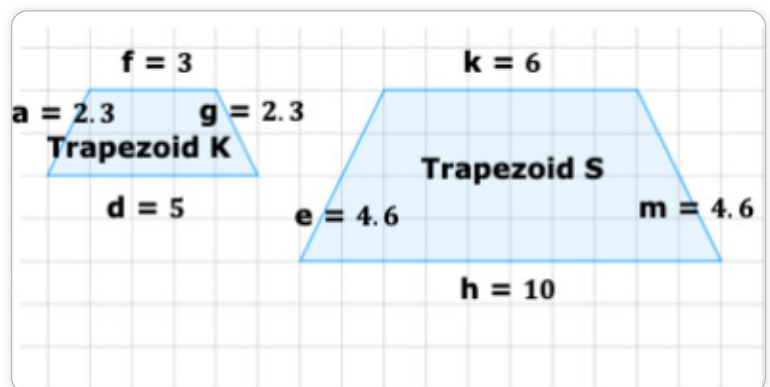
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

5.6.2 Exploration gives students an opportunity to work together to discover the relationship between scale factor and area through patterns and reasoning. The questions within the workbook provide the scaffolding to foster the discussion.

DIFFERENTIATE INSTRUCTION

During 5.6.3 Guided Instruction, consider whether students are able to complete the table relating the triangles using different scale factors on their own, in small groups, or with teacher support. Consider also differentiating by providing additional scale factors for students to practice within this component based on their level of readiness.



5.6.1 WARM-UP: WORLD'S LARGEST GUMMY BEAR

TEACHER MOVES

Consider sharing student responses anonymously or creating your own possible responses in order to discuss how to create a reasonable versus unreasonable estimate of scale factor.



5.6.1 Warm-Up: World's Largest Gummy Bear

The World's Largest Gummy Bear is the equivalent of 1,400 regular sized gummy bears and packs a whopping 6,120 calories.



1. What do you wonder about the World's Largest Gummy Bear?

Answers will vary. I wonder whether the taste has been concentrated or diluted. I wonder how much it weighs.

2. The height of the World's Largest Gummy Bear is $9\frac{1}{2}$ inches (in.). Guess what the scale factor for the World's Largest Gummy Bear is to a normal gummy bear.

Answers will vary. I know that the gummy bear is the size of 1,400 gummy bears. I also know that one gummy bear is about $\frac{1}{2}$ inch. An estimate that I have is that the scale factor would be 19.

3. The height of a regular gummy bear is 0.8 in. Use your scale factor to determine the accuracy of your guess.

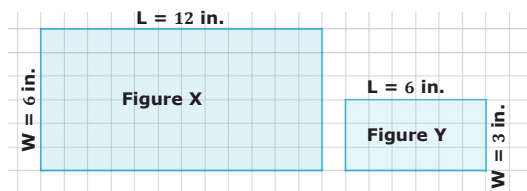
Answers will vary. 0.8 is about one and a half times what I had guessed it was. Given this, I would adjust my scale factor to 12 because $0.8 \times 12 = 9.6$.



5.6.2 Exploration: Scale Factor and Area

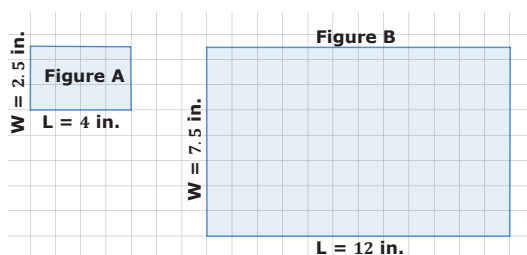
Work with your partner to complete the following.

- The same figures from Paolo's advertisement are shown where Figure X has a length (L) of 12 units and a width (W) of 6 units, and Figure Y has a length (L) of 6 units and a width (W) of 3 units.



- The scale factor from Figure X to Figure Y is $\frac{1}{2}$.
- What is the area of Figure X?
 $6 \times 12 = 72 \text{ units}^2$
- What is the area of Figure Y?
 $3 \times 6 = 18 \text{ units}^2$
- What is the ratio of the area of Figure Y to the area of Figure X?
 $18:72$ or $1:4$ or $\frac{1}{4}$
- What relationship, if any, is there between the scale factor and the ratio of the areas?
The ratio of the areas appears to be the square of the scale factor. $(\frac{1}{2})^2 = \frac{1}{4}$

- Figure B is a scale model of Figure A. Figure B has a length (L) of 12 inches (in.) and a width (W) of 7.5 in. Figure A has a length (L) of 4 in. and a width (W) of 2.5 in.



- What is the scale factor from Figure A to Figure B?
The ratio is 2.5:7.5 or 1:3. The scale factor from Figure A to Figure B is 3.
- What is the area of Figure A?
 $2.5 \text{ in} \times 4 \text{ in} = 10 \text{ in}^2$
- What is the area of Figure B?
 $7.5 \text{ in} \times 12 \text{ in} = 90 \text{ in}^2$
- How does the scale factor from Figure A to Figure B relate to the ratio of the area of Figure A to Figure B?
The ratio of the areas is 10:90 or 1:9 and 9 is the square of the scale factor, $3^2 = 9$.
- Discuss the relationship between the scale factor and the area you observed in problems 1 and 2.
The square of the scale factor can be used to find the area of the scaled figure by multiplying the square of the scale factor and the area of original figure.

5.6.2 EXPLORATION: SCALE FACTOR AND AREA

TEACHER MOVES

Some students may not be able to make the connection between scale factor and area after only one example. If students are not able to verbalize a clear relationship in question 1e, that is reasonable at this point. Students should continue the Exploration and may notice the relationship after question 2. Ensure they have the time to explore before interjecting.

ENRICHMENT

Students may struggle to explain the relationship between scale factor and area. Ask questions to draw connections to prior knowledge of scale factor and area:

- What is the relationship between scale factor and perimeter?
- What units is area measured in? (square units)
- What is the ratio of the areas?
- How does that relate to the original scale factor?

5.6.2 EXPLORATION: SCALE FACTOR AND AREA (CONTINUED)

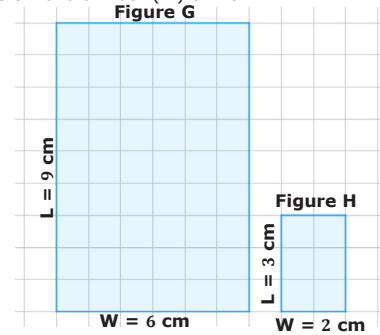
DIFFERENTIATE INSTRUCTION

Consider using this opportunity to work with some small groups that may be struggling, if needed. Ask them to predict the area if they've been given the scale factor and the dimensions of the first shape.

TEACHER MOVES

Consider having a few students share out their conjectures with the whole group before moving to the next component. Hearing other students' ideas may spark ideas for students who haven't been able to make the connection yet.

3. Figure G has a length (L) of 9 centimeters (cm) and a width (W) of 6 cm. Figure G was scaled down by a factor of $\frac{1}{3}$ to create Figure H. Figure H has a length (L) of 3 cm and a width (W) of 2 cm.

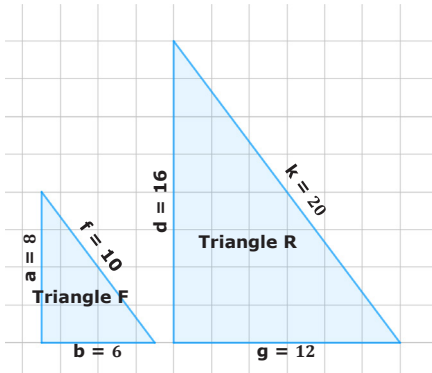


- Find the area of Figure G.
 $9\text{ cm} \times 6\text{ cm} = 54\text{ cm}^2$
 - Without finding the product of the sides, what do you expect the area of Figure H to be?
I would expect the area to be $\frac{1}{9}$ the area of Figure G or 6 cm^2 . $\frac{1}{9} \times 54 = 6$
 - Verify your answer by finding the product of the sides of Figure H.
 $3\text{ cm} \times 2\text{ cm} = 6\text{ cm}^2$
4. With your partner, write a rule that can be used to calculate the area of a scale model if given only an original area and a scale factor.
Multiply the area of the original figure by the scale factor squared.



5.6.3 Guided Instruction: Comparing Perimeter and Area of Scaled Triangles and Trapezoids

- Right Triangle F has a base (b) of 6 units, a height (a) of 8 units, and a third side (f) of 10 units. Right Triangle R is a scaled image of Triangle F. Triangle R has a base (g) of 12 units, a height (d) of 16 units, and a third side (k) of 20 units.



- The scale factor from Triangle F to Triangle R is 2.
- Compare the perimeter of Triangle F to the perimeter of Triangle R.
Triangle F perimeter: $8 + 10 + 6 = 24$ units
Triangle R perimeter: $20 + 12 + 16 = 48$ units
The perimeter of Triangle R is 2 times the perimeter of Triangle F.

- Using Triangle F and the listed scale factors, complete the table.

Scale Factor	Base (cm)	Height (cm)	3 rd side (cm)	Perimeter (cm)	Area (cm ²)
1	6	8	10	24	24
3	18	24	30	72	216
$\frac{1}{2}$	3	4	5	12	6

- Discuss with your partner whether the rules you wrote for calculating the perimeter or area of a scaled rectangle, given a scale factor and the original perimeter or area, apply to triangles.
The rule generated for the area and perimeter of scaled rectangles also applies to scaled triangles. The scale factor relating the figures can be used to relate the perimeters. The square of scale factor can be used to relate the areas.

5.6.3 GUIDED INSTRUCTION: COMPARING PERIMETER AND AREA OF SCALED TRIANGLES AND TRAPEZOIDS

TEACHER MOVES

Notice and Wonder

Consider implementing a Notice and Wonder routine as students consider the figures in question 1. Encourage students to wonder if other scaled figures will have the same relationships related to scale factor, perimeter, and area.

TEACHER MOVES

After students complete the table, take time to stop and discuss how scale factor affects the resulting figures.

- If a scale factor is between 0 and 1, do the original measurements become smaller or larger? And what if the scale factor is greater than 1?
- Will this always be true? How do you know?
- What operation is used when applying scale factor? (Multiplication, even if the scale is getting smaller. You will multiply by a fraction.)
- Why do you think the scale factor is squared when applied to area but not perimeter?

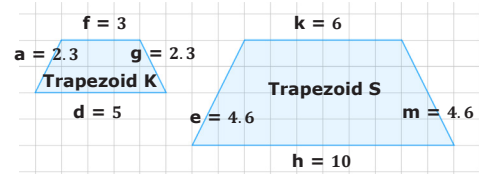
5.6.3 GUIDED INSTRUCTION: COMPARING PERIMETER AND AREA OF SCALED TRIANGLES AND TRAPEZOIDS (CONTINUED)

TEACHER MOVES

Notice and Wonder

Consider implementing a Notice and Wonder routine as students consider the figures in question 2. Encourage students to wonder if other scaled figures will have the same relationships related to scale factor, perimeter, and area.

2. Trapezoid K has two bases. One base (d) is 5 units long and the other 3 (f) units long. The other two sides (a and g) have equal lengths of 2.3 units. A scale factor of 2 has been applied to Trapezoid K resulting in Trapezoid S. Trapezoid S has two bases, one (h) 10 units long and the other (k) 6 units long. The other two sides (e and m) have equal lengths of 4.6 units.



- How does the perimeter of Trapezoid K compare to the perimeter of Trapezoid S?
Trapezoid K perimeter: 12.6 units
Trapezoid S perimeter: 25.2 units
The perimeter of Trapezoid S is 2 times the perimeter of Trapezoid K.
- Explain how these perimeters relate to the scale factor.
The perimeter of the scaled trapezoid, Trapezoid S, is equivalent to the scale factor times the perimeter of the original trapezoid, Trapezoid K.
- How should the areas of the figures relate?
The area of the scaled trapezoid, Trapezoid S, should be 4 times the area of the original trapezoid, Trapezoid K, because they should relate by the square of the scale factor.



5.6.4 Wrap-Up: Growth Mindset

1. Complete the table as you reflect on today’s lesson.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to “un-stick” myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always

Answers will vary.

5.6.4 WRAP-UP: GROWTH MINDSET

DIFFERENTIATE INSTRUCTION

Consider displaying an anchor chart with examples and nonexamples of each mindset for students to reference as they reflect. This will aid students in learning to monitor their own behavior as they work toward understanding their own progress and areas for improvement.